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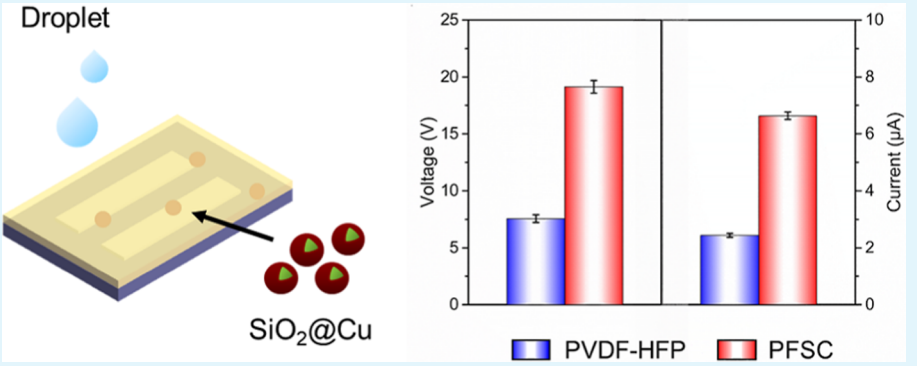
Enhanced Liquid–Solid Triboelectric Nanogenerators with SiO_2@Cu-Doped Composite Coating for Droplet Energy Harvesting

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ABSTRACT: Rainwater is a clean and abundant energy source, and the liquid−solid triboelectric nanogenerator (LS-TENG) has emerged as a promising technology for harvesting and converting raindrop energy into electricity. Despite its potential, practical applications of LS-TENG have been limited by its low output performance and complex fabrication processes. Herein, we propose a simple and efficient strategy for constructing high-performance LS-TENG by introducing a novel composite triboelectric layer composed of poly(vinylidene fluoride-*co*-hexafluoropropylene) (PVDF-HFP) doped with SiO_2@Cu and 1H,1H,2H,2H-perfluorooctyltriethoxysilane. This design simultaneously enhances the dielectric constant and hydrophobicity of the triboelectric coating through a polarization effect and surface tuning. As a result, the LS-TENG achieves an excellent output voltage of 19 V and a current of 6.7 μA from a single waterdrop, corresponding to 2.5 and 2.8 times those of undoped PVDF-HFP, and can successfully power small electronic devices. Furthermore, the entire LS-TENG, including both the triboelectric and electrode layers, is fabricated via facile, in situ coating processes, offering good scalability, substrate adaptability, and optical transparency. Therefore, this work integrates material-level output enhancement with process-level simplicity, holding great potential for practical applications in droplet energy harvesting.

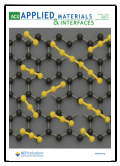
KEYWORDS: *droplet energy, composite coating, liquid−solid triboelectric nanogenerator, dielectric constant, hydrophobicity*

1. INTRODUCTION

Nonrenewable energy resources such as fossil fuels contribute significantly to environmental issues.[1,2] To alleviate the problems of global energy shortage and environmental pollution, various renewable energy resources, including solar, wind, ocean, and raindrop energy, have been explored in recent years.[3−5] Among them, harvesting raindrop energy is of great significance in areas with abundant rainfall. Traditionally, hydro-electromagnetic generators have been used to capture this energy.[6] However, large-scale implementation remains challenging due to the complexity and cost of facilities as well as the low frequency and widespread distribution of raindrops. In 2014, Wang et al. introduced the liquid−solid triboelectric nanogenerator (LS-TENG), capable of converting droplet energy into electricity through frictional electrification and electrostatic induction between droplets and insulating materials.[7] With its simple structure, broad material selection, and efficient low-frequency conversion, LS-TENG has emerged as a promising method for harvesting raindrop energy.[8−12] However, the intentional use of water as the positive triboelectric material restricts its electrical output performance compared with solid−solid triboelectric nanogenerators.

The electrical output performance of the LS-TENG is closely related to the intrinsic properties of the triboelectric materials. Polyvinylidene fluoride and its copolymers, such as poly(vinylidene fluoride-*co*-hexafluoropropylene) (PVDF-

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HFP), are commonly employed as negative triboelectric materials due to their excellent electronegativity and processability.[13,14] Nevertheless, further improvement of the electrical output performance remains imperative. Recent advances have focused on improving the electronegativity, dielectric property, and hydrophobicity of the triboelectric materials through various modification techniques.[15−19] For example, Du et al. fabricated a strongly hydrophobic triboelectric layer featuring a nanoscale surface structure.[20] Choi et al. effectively modified the surface potential by injecting electrical charges.[21] Despite significant efforts to enhance the electrical output performance of the LS-TENG, there remains a need for a more efficient and scalable approach with a simple fabrication process and minimal equipment.

The construction of composite material through nanoparticle doping via a simple coating process is considered a suitable approach for easily altering the intrinsic properties of materials.[22−26] Various types of nanoparticles are used as fillers to enhance composite coatings with excellent charge capture and storage abilities. These include inorganic oxide nanoparticles, such as SiO_2, TiO_2, and $BaTiO_3$, and conductive nanofillers like Ag, Au, and carbon nanotubes.[27−32] Furthermore, coating techniques such as spraying, spinning, drop-casting, and dip-coating offer scalable and in situ fabrication, making them highly adaptable for practical and large-scale applications.[33−36] Therefore, in the field of LS-TENG for droplet energy harvesting, utilizing composite materials is significant, especially for exploring the combined effects of fillers to enhance electrical output. Additionally, maximizing the potential benefits of coating technology across all layers of the LS-TENG is equally important.

This study presents a composite coating by incorporating SiO_2@Cu and 1H,1H,2H,2H-perfluorooctyltriethoxysilane (FOTS) into a PVDF-HFP matrix, which serves as the negative triboelectric material in LS-TENG. The addition of SiO_2@Cu and FOTS simultaneously enhances the dielectric constant and hydrophobicity of the coating through the polarization effect and surface tuning. Therefore, the electrical output performance of the LS-TENG is significantly enhanced with the PVDF-HFP/FOTS/SiO_2@Cu (PFSC) composite coating. To achieve optimal performance, the influences of composition variables such as composition ratio and doping concentration as well as external factors like inclined angle and dropping height are systemically investigated. Under the optimized conditions, LS-TENG generates an astonishing output voltage of 19 V and a current of 6.7 μA from a single waterdrop, representing increases of 2.5 and 2.8 times, respectively, compared to pristine PVDF-HFP. This output is sufficient to illuminate 27 light-emitting diodes (LEDs) and power a timer. Additionally, all layers in the LS-TENG are fabricated using a simple coating process with drop-casting PFSC as a triboelectric layer and spraying silver nanowires (AgNWs) as a conductive layer. This method offers great advantages in terms of high scalability, design flexibility, and transparency. Consequently, this work develops a high-performance LS-TENG with an easy fabrication process based on a PFSC composite coating, providing a promising approach for harvesting raindrop energy and holding great potential for powering electronic devices effectively.

2. EXPERIMENTAL SECTION

2.1. Materials. SiO_2 nanospheres (300 nm) were purchased from Suzhou Knowledge & Benefit Sphere Tech. Co., Ltd. Hydrazine hydrate ($N_2H_4{\cdot}H_2O$, 98%), polyethylene glycol (PEG-400), poly(vinylidene fluoride-*co*-hexafluoropropylene) (PVDF-HFP, M_w = 455,000), and 1H,1H,2H,2H-perfluorooctyltriethoxysilane (FOTS, 97%) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Copper sulfate pentahydrate ($CuSO_4{\cdot}5H_2O$, 99%), sodium hydroxide (NaOH, 96%), *N*,*N*-dimethylformamide (DMF, 99.5%), and anhydrous ethanol (99.7%) were purchased from Sinopharm Chemical Reagent Co, Ltd. Silver nanowires (AgNWs, d = 30 nm, l = 20 μm, 2 mg/mL in ethanol) were purchased from Jiangsu XFNANO Materials Tech Co., Ltd.

2.2. Synthesis of SiO_2@Cu. To prepare the suspension, 0.2 g of SiO_2, 0.5 g of $CuSO_4{\cdot}5H_2O$, 100 mL of deionized water, and 0.5 mL of PEG-400 were added in a glass beaker and stirred at 70 °C for 30 min. Meanwhile, 0.6 mL of $N_2H_4{\cdot}H_2O$ and 50 mL of deionized water were mixed in another glass beaker and stirred under the same conditions. The pH of the first suspension was then adjusted to 11 with a NaOH solution. Both solutions were subsequently mixed and stirred for another 30 min. Finally, the resulting suspension was centrifuged, and the precipitate was washed several times and dried in a vacuum chamber. For varying Cu to SiO_2 ratios, different amounts of $CuSO_4{\cdot}5H_2O$ (0.20, 0.50, 1.25, and 3.13 g) were added.

2.3. Preparation of PFSC Suspension. Initially, 1.4 mg of as-synthesized SiO_2@Cu and 2.7 μL of FOTS were ultrasonically dispersed in 700 μL of DMF for 1 h. Subsequently, 70 mg of PVDF-HFP was dissolved in the resulting suspension and magnetically stirred for 6 h. For suspensions with different doping concentrations, different amounts of SiO_2@Cu (0.7, 1.4, 2.1, and 2.8 mg) were used.

2.4. Fabrication of PFSC LS-TENG. The poly(methyl methacrylate) (PMMA) substrate was first cleaned with ethanol and treated with an oxygen plasma. To construct the double-electrode conductive coating, 800 μL of AgNW dispersion was uniformly sprayed onto the substrate through a 0.5 mm nozzle from a distance of 100 mm, with the substrate covered by a mask. The resulting coating was then annealed at 80 °C for 2 h to evaporate the solvent and fuse the nanowires. Two short copper tapes were attached as connection wires for the electrical measurements. Thereafter, the prepared PFSC suspension was drop-cast to create the triboelectric coating, which was dried at 60 °C for 3 h.

2.5. Characterizations and Measurements. The morphologies of nanoparticles and coatings were characterized by a field emission scanning electron microscope (Zeiss Ultra 55, Germany) with an energy dispersion spectrometer, and a transmission electron microscope (HT7800, Japan). The chemical compositions of nanoparticles and coatings were characterized by X-ray diffraction (XRD, Rigaku SmartLab 9 kW, Japan) from 10° to 80°, X-ray photoelectron spectroscopy (XPS, Thermofisher Nexsa, USA), and Fourier transform infrared spectroscopy (FTIR, Thermofisher Nicolet 6700, USA). The dielectric constants were tested by a dielectric spectrometer (Novocontrol Technologies BDS 50, USA) with Faraday cage shielding in parallel-plate capacitor geometry within the frequency range from 10^3 Hz to 10^7 Hz. The water contact angles (4 μL), sliding angles, and contact angle hysteresis (12 μL) were tested by a contact angle meter (Dataphysics OCA 25, Germany), and the droplet movement states were tested by a high-speed camera (Rocketech Acuteye-1M-2000, China). The light transmittances were tested by an ultraviolet−visible spectrophotometer (PerkinElmer Lambda 750, USA) in the visible range from 380 to 780 nm. The leakage current density was measured by an electrometer (Keithley 6517B, USA). The output voltage and short-circuit current of the LS-TENG were measured by an electrometer (Keithley 6517b, USA) and a digital multimeter (Keithley 6500, USA). Water droplets with a controlled volume of 30 μL were produced with a syringe pump (Leienyl LINZ-9A, China).

3. RESULTS AND DISCUSSION

3.1. Construction of PFSC Coating. According to pioneering research, the droplet energy harvesting mechanism in a double-electrode mode for LS-TENG is shown in Figure 1a.[8,15,21] When a droplet makes contact with the triboelectric

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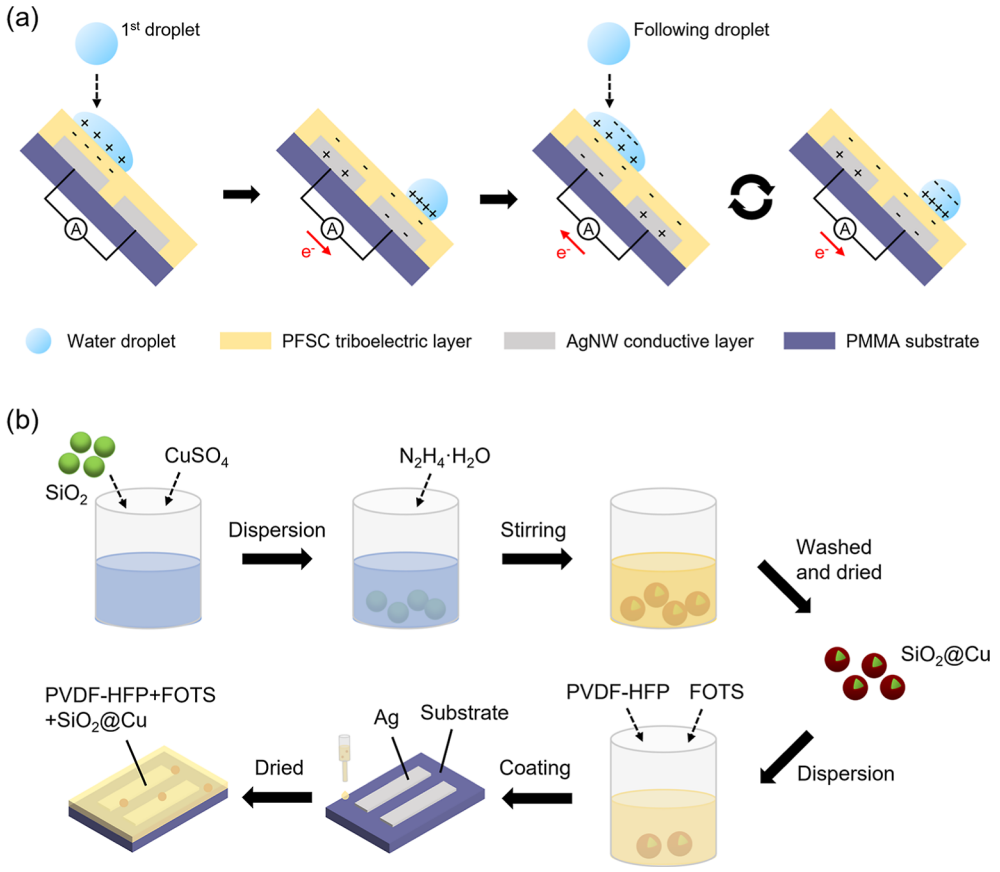


Figure 1. Working mechanism and construction method of LS-TENG with PFSC triboelectric coating. (a) Schematic diagram for the working mechanism of LS-TENG. (b) Schematic diagram for the construction method of SiO_2@Cu and PFSC coating.

layer, frictional electrification occurs due to differences in electron affinity, resulting in the droplet gaining a positive charge, while the triboelectric layer acquires a negative charge. As the droplet moves downward, the negative charges on the triboelectric layer are retained. Electrostatic induction is generated between the triboelectric and conductive layers to meet electrical neutrality, leading to the migration of free electrons from the first electrode to the second electrode. Once this process reaches saturation, contact of a subsequent droplet with the triboelectric layer creates a double electric layer, with negative charges on the triboelectric layer being partly screened. This interaction prompts the movement of free electrons from the second electrode back to the first electrode, restoring the electrical equilibrium. As the droplet slides down, screened negative charges are recovered, and electrons move in the opposite direction again. This mechanism indicates that the intrinsic properties of the triboelectric layer impact the quantity of transferred and induced electrons, thereby influencing the electrical output performance of the LS-TENG. Therefore, a PFSC composite coating with a superior dielectric property and hydrophobicity is designed.

The synthesis of SiO_2@Cu and construction of the PFSC composite coating are illustrated in Figure 1b. SiO_2@Cu is prepared by depositing Cu on the surface of SiO_2 nanospheres through an in situ reduction reaction involving $CuSO_4 \cdot 5H_2O$ and $N_2H_4 \cdot H_2O$. This wet-chemical synthesis method eliminates the need for specialized equipment, enabling scalable production and storage for multibatch applications. Subsequently, SiO_2@Cu and FOTS are incorporated into PVDF-HFP, and the resulting uniformly mixed suspension is drop-cast to fabricate the PFSC coating as the negative triboelectric layer for LS-TENG. The solution-based coating technique operates without sophisticated instrumentation, combining fundamental simplicity with robust scalability for manufacturing.

Figure 2a,b shows the scanning electron microscopy (SEM) images of SiO_2 and SiO_2@Cu. The comparison reveals that Cu is successfully deposited on the SiO_2 surface. Figure S1 shows the transmission electron microscopy (TEM) images of SiO_2 and SiO_2@Cu, further corroborating the core−satellite structure. Figure 2c displays the XRD patterns of SiO_2, Cu, and SiO_2@Cu. The peaks at 2θ values of 43.3°, 50.4°, and 74.1° correspond to the (111), (200), and (220) planes of cubic Cu, respectively. The broad peak at approximately 22.9° indicates the amorphous structure of SiO_2. These results confirm the presence of crystalline Cu on the surface of the amorphous SiO_2 surface. To further verify the valence state of Cu, the XPS spectrum of SiO_2@Cu is depicted in Figure S2. The peaks at 934 and 954 eV are attributed to Cu $2p_{3/2}$ and Cu $2p_{1/2}$, respectively, with minimal presence of satellite peaks. These observations suggest that Cu in SiO_2@Cu predominantly exhibits a zero-valence state. In addition, Figure S3 shows the energy-dispersive spectroscopy (EDS) mappings, revealing that the mass ratio of Cu to SiO_2 in SiO_2@Cu varies between approximately 1:4, 2:3, 3:2, and 4:1.

Figure 2d depicts the SEM image and EDS mappings of the PFSC composite coating. The existence and relatively uniform distribution of Cu, Si, and O elements validate the successful and even effective doping of SiO_2@Cu within the composite coating. To further confirm the composition, Figure S4

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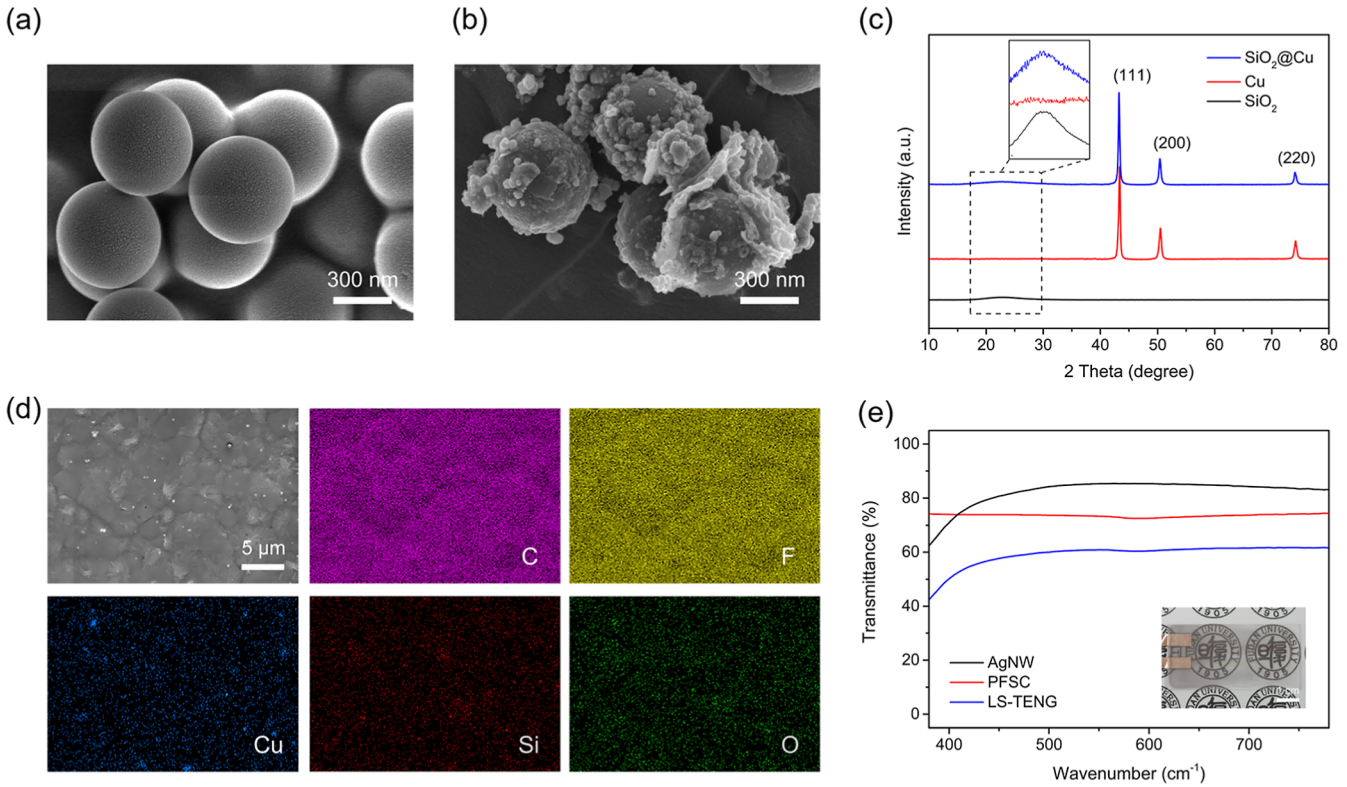


Figure 2. Characterizations of SiO_2@Cu and PFSC coating. SEM images of (a) SiO_2 and (b) SiO_2@Cu. (c) XRD pattern of SiO_2, Cu, and SiO_2@Cu. (d) SEM image and EDS mappings of PFSC coating. (e) Light transmittance of AgNW conductive coating, PFSC triboelectric coating, and LS-TENG (optical image is shown in the inset).

presents the XRD pattern and FTIR spectra of PVDF-HFP, PVDF-HFP/FOTS (PF), PVDF-HFP/FOTS/SiO_2 (PFS), PVDF-HFP/FOTS/Cu (PFC), and PFSC coatings. The XRD pattern demonstrates a peak at 43.5°, corresponding to the (111) plane of Cu, thereby confirming the existence of the Cu. In the FTIR spectrum, a weak peak at 1105 cm^{-1} is observed, corresponding to the characteristic stretching vibration of Si−O in SiO_2, which confirms the presence of SiO_2. The thickness of the PFSC coating, depicted in Figure S5, is approximately 19 μm. Furthermore, Figure S6 displays the SEM images of PFSC coatings with varying SiO_2@Cu doping concentrations. As the doping concentration increases to 3 and 4 wt %, there is a greater exposure of SiO_2@Cu on the surface, leading to an increase in agglomeration.

Besides, as depicted in Figure 2e, the AgNW conductive layer, PFSC triboelectric layer, and the entire LS-TENG exhibit the highest light transmittance of 85%, 73%, and 61%, respectively, within a visible wavelength range of 380−780 nm, at an AgNW spraying concentration of 0.5 mg/mL. These results demonstrate the exceptional transparency of the obtained coatings and LS-TENG.

3.2. Output Performance of LS-TENG. To investigate the impact of fillers on the electrical output enhancement of the LS-TENG, control tests were conducted using triboelectric coatings with different compositions, as shown in Figure 3a−c. The output voltage of LS-TENG increases progressively with the use of PVDF-HFP, PF, PFS, PFC, and PFSC coatings, rising from 7.6 V, 11 V, 13 V, and 16 V to 19 V. Similarly, the short-circuit current rises from 2.4 μA, 3.9 μA, 4.9 μA, and 5.6 μA to 6.7 μA. The detailed current−time curves for each coating are presented in Figure S7, from which the corresponding integrated charges are derived. A clear increasing trend is observed, with values rising from 6.4, 9.7, 12, and 13 nC to 16 nC. These results confirm that the LS-TENG with PFSC composite coating exhibits the highest electrical output, with its voltage, current, and charge reaching approximately 2.5−2.8 times those of the device based on pristine PVDF-HFP. Furthermore, a performance comparison summarized in Table S1 reveals that the PFSC-based LS-TENG achieves comparable or superior output to many previously reported LS-TENGs, despite using a smaller droplet volume and a simple fabrication process.

Next, to further determine the optimal composition for the triboelectric coating of LS-TENG, a series of experiments were carried out, varying the composition ratios of SiO_2@Cu, doping concentrations of SiO_2@Cu and FOTS, and drop-casting volumes of the PFSC suspension. Figures 3d,e and S8a show the electrical output of LS-TENG at different SiO_2@Cu ratios under constant conditions: 2 wt % for SiO_2@Cu doping, 5 wt % for FOTS doping, and 700 μL PFSC drop-casting volume. As the mass ratio of Cu to SiO_2 increases from 1:4 to 2:3, the electrical output increases from 14 V, 4.8 μA, and 13 nC to 19 V, 6.7 μA, and 16 nC. However, further increases in the mass ratio to 3:2 and 4:1 result in a decline in output to 17 V, 5.9 μA, and 15 nC and 16 V, 5.4 μA, and 14 nC, respectively. Therefore, the optimal performance is achieved at a Cu-to-SiO_2 mass ratio of 2:3. Figures 3f,g and S8b show the electrical output at different SiO_2@Cu doping concentrations under this optimal mass ratio. As the doping concentration increases from 1 to 2 wt %, the electrical output correspondingly rises from 16 V, 5.3 μA, and 13 nC to 19 V, 6.7 μA, and 16 nC. Further increases to 3 and 4 wt % cause a decline in output to 18 V, 6.0 μA, and 15 nC and 16 V, 4.7

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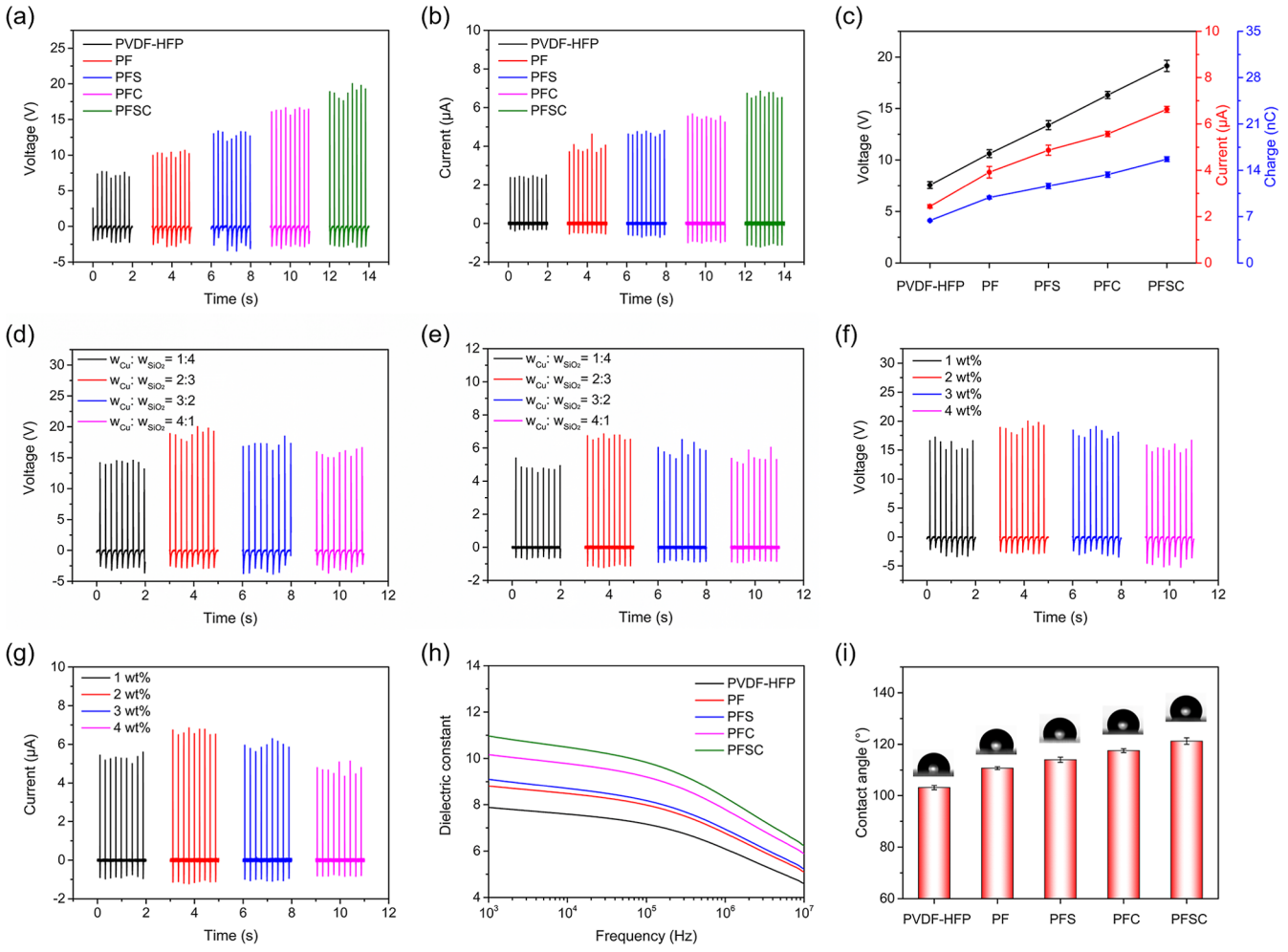


Figure 3. Output performance of LS-TENG based on triboelectric coating with different compositions and its enhancement mechanism. (a) Output voltage, (b) short-circuit current, and (c) integrated charge of LS-TENG with PVDF-HFP, PF, PFS, PFC, and PFSC coatings. (d) Output voltage and (e) short-circuit current of LS-TENG at different SiO_2@Cu composition ratios. (f) Output voltage and (g) short-circuit current of LS-TENG at different SiO_2@Cu doping concentrations. (h) Dielectric constant of PVDF-HFP, PF, PFS, PFC, and PFSC coatings. (i) Water contact angle of PVDF-HFP, PF, PFS, PFC, and PFSC coatings.

μA, and 12 nC, respectively. Therefore, the ideal doping concentration for SiO_2@Cu is determined to be 2 wt %.

In addition, Figure S9 supplements the electrical output performance at different FOTS doping concentrations. As the concentration rises from 2.5 to 5 wt %, the electrical output increases rapidly. However, beyond 5 wt %, the increase becomes less pronounced. Excessive addition of fluorine modifier has little effect on improving electrical output and poses environmental concerns. Thus, a doping concentration of 5 wt % is considered optimal. Furthermore, the variation in drop-casting volumes of the PFSC solution directly impacts the thickness of the triboelectric coating, which in turn influences the electrical output performance of LS-TENG. As shown in Figure S10, the electrical output decreases with increasing drop-casting volume. Therefore, 700 μL is selected as the suitable drop-casting volume.

Both the dielectric constant and hydrophobicity of the triboelectric layer critically influence the electrical output performance of LS-TENG. Figure 3h displays the dielectric constants of different coatings across the frequency range of 10^3–10^7 Hz. Consistent with previous studies, the dielectric constant decreases with increasing frequency due to the lag of polarization response to the alternating electric field.[25] At 10^3 Hz, as summarized in Figure S11, the dielectric constants of PVDF-HFP, PF, PFS, PFC, and PFSC coatings are approximately 7.8, 8.6, 8.9, 10, and 11, respectively. This enhancement arises from synergistic contributions of all of the dopants. FOTS boosts molecular polarization through its fluorinated groups.[26,37,38] SiO_2 induces interfacial polarization via the Maxwell–Wagner–Sillars effect.[39,40] Cu contributes to interfacial polarization and promotes local microcapacitor formation due to its high conductivity.[41,42] SiO_2@Cu's core–satellite architecture further enhances these effects by introducing abundant heterogeneous interfaces.[25,43] Therefore, the PFSC composite coating exhibits the highest dielectric constant, reaching 141% of that of pure PVDF-HFP.

The dielectric constant is quantitatively correlated with surface charge density and electrical output performance.[37] Considering the triboelectric layer as a capacitor, the transferred charge ($Q(t)$) is given by

$$Q(t) = CV(1 - e^{-t/RC}) \quad (1)$$

where C is the capacitance of the triboelectric layer, V is the applied voltage, t is the charging time, and R is the resistance. Assuming the voltage is constant, the capacitance is given by

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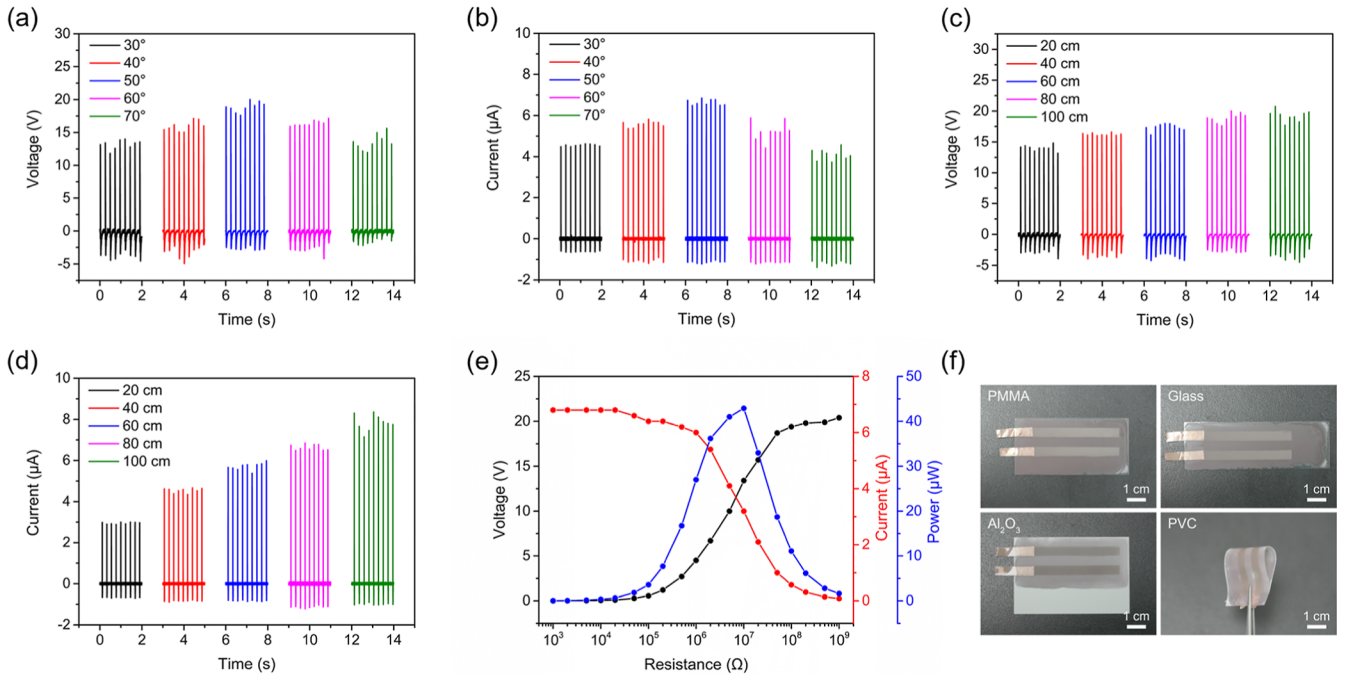


Figure 4. Output performance of LS-TENG based on PFSC triboelectric coating with various external conditions. (a) Output voltage and (b) short-circuit current of LS-TENG at different inclined angles. (c) Output voltage and (d) short-circuit current of LS-TENG at different dropping heights. (e) Output voltage, current, and power of LS-TENG with different load resistances. (f) Photographs of LS-TENG on different substrates.

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} \tag{2}$$

where ε_0 is the vacuum dielectric constant, ε_r is the dielectric constant of the triboelectric layer, A is the effective contact area, and d is the layer thickness. Once charge saturation is reached, the transferred charge can be approximated as $Q \approx CV$. Accordingly, the surface charge density (σ) is expressed as

$$\sigma = \frac{Q}{A} = \frac{\varepsilon_0 \varepsilon_r V}{d} \tag{3}$$

Therefore, a higher dielectric constant leads to a greater surface charge density. The increased surface charge density ultimately improves both the voltage and current output of LS-TENG. Detailed derivations are provided in Note S1.

Hydrophobicity is another critical factor influencing the electrical output performance of LS-TENG. As shown in Figure 3i, the water contact angle increases from 103° for PVDF-HFP to 122° for PFSC, with intermediate values of 111°, 114°, and 118° for PF, PFS, and PFC, respectively. This enhancement arises from both surface energy reduction and morphological modification. The incorporation of fluorinated FOTS effectively lowers the surface energy.[44] Meanwhile, the embedded SiO_2@Cu nanoparticles induce microstructural surface protrusions, which facilitate air entrapment at the solid–liquid interface and form a mixed solid–liquid–air interface, ultimately enhancing the contact angle.[45] Consequently, the PFSC composite coating exhibits the highest hydrophobicity. Furthermore, sliding angle and contact angle hysteresis are shown in Figure S12, both exhibiting a decreasing trend with increasing hydrophobicity.

Enhanced hydrophobicity facilitates droplet movement and reduces water residue, thereby accelerating contact-separation dynamics and mitigating charge screening, both of which are essential for optimizing electrical output. This correlation is supported by high-speed images and dynamic analyses presented in Figures S13–S15. Specifically, $\Delta A/\Delta t$, representing the rate of solid–liquid contact area variation during the droplet contact-separation process, increases with improved hydrophobicity. This parameter serves as an indicator of droplet dynamics and exhibits a positive correlation with electrical output, as previously demonstrated.[46] Concurrently, the water residue is significantly reduced as hydrophobicity increases, with the PFSC coating showing nearly negligible residue. This promotes more effective contact and minimizes charge shielding during successive droplet interactions.[47] Collectively, these findings confirm that enhanced hydrophobicity of the PFSC composite coating improves the electrical output performance of LS-TENG.

Figure 4 presents the influence of various external conditions, such as inclined angle, dropping height, load resistance, and substrate, on the electrical output performance of the LS-TENG with a PFSC triboelectric coating. Figures 4a,b and S16a show the electrical output performance of LS-TENG at different inclined angles, maintaining a constant dropping height of 80 cm. As the inclined angle increases from 30° to 50°, the electrical output correspondingly rises from 14 V, 4.5 μA, and 12 nC to 19 V, 6.7 μA, and 16 nC. However, further increases lead to a decline in output. This result is ascribed to electrostatic shielding caused by the droplet residue at smaller inclinations and a reduction in the effective contact area between the droplet and the triboelectric coating at larger inclinations. In addition, Figures 4c,d and S16b show the electrical output performance of the LS-TENG at various dropping heights with an optimal inclination of 50°. As the dropping height is increased from 20 to 100 cm, the electrical output progressively rises from 14 V, 3.0 μA, and 11 nC to 19 V, 8.1 μA, and 16 nC. This enhancement is mainly attributed to the greater drop velocity and impact force, making the contact-separation process more sufficient. However, excessively high dropping heights can cause droplet splashing,

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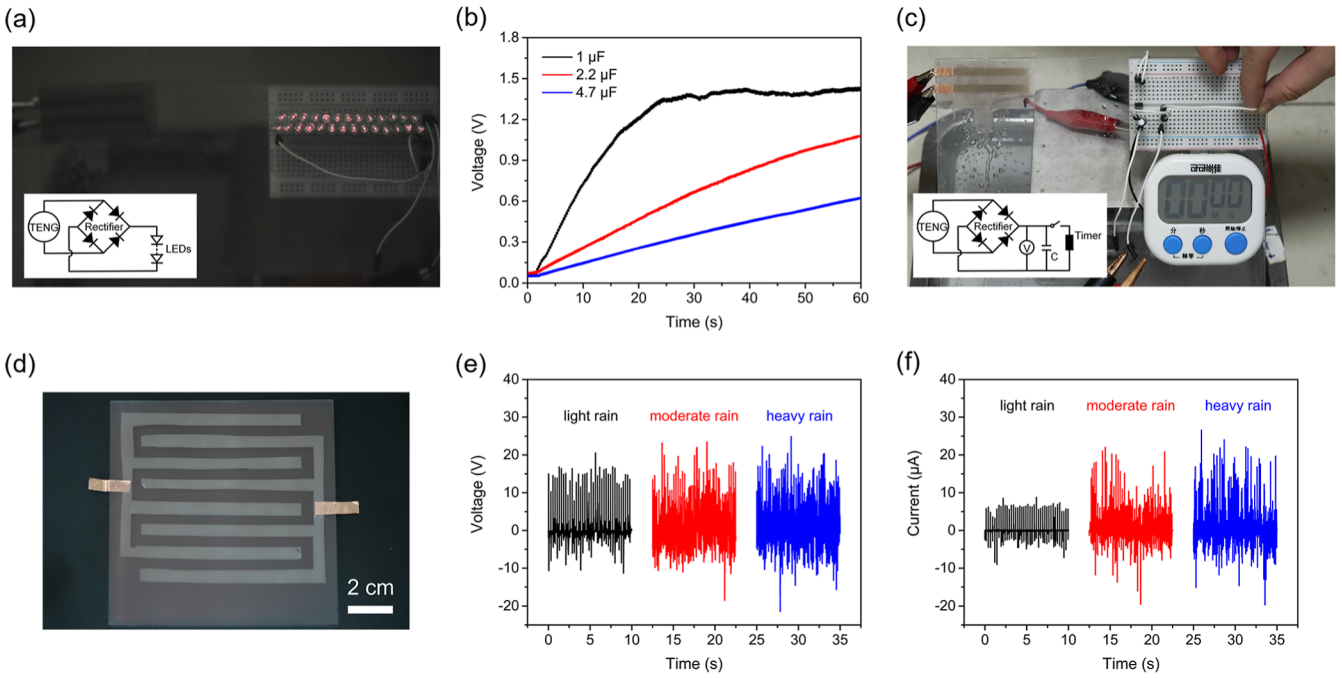


Figure 5. Applications of LS-TENG with PFSC triboelectric coating. (a) Photograph of LS-TENG lighting up LEDs (the circuit is shown in the inset). (b) Charging curve of different capacitors. (c) Photograph of LS-TENG powering a timer (the circuit is shown in the inset). (d) Photograph of large-scale LS-TENG with interdigital electrodes. (e) Output voltage and (f) short-circuit current of large-scale LS-TENG under different rainfall modes.

resulting in relatively unstable output, especially in terms of the current. The corresponding Weber number is also calculated by $We = \frac{\rho v_n^2 D}{\sigma}$, where ρ is the water density, v_n is the normal velocity, D is the diameter, and σ is the surface tension (Figure S17). Hence, a height of 80 cm is chosen for the subsequent experiments.

Besides, to explore the impact of load resistance and identify the optimal match, output voltage, current, and power measurements of the LS-TENG are taken across a resistance range from 10^3 to 10^9 Ω, as shown in Figure 4e. With the increase in resistance, the voltage rises correspondingly, whereas the current decreases. The power initially increases and then decreases, reaching a maximum of 43 μW when the resistance is 1×10^7 Ω. As shown in Figure S18, the PFSC-based LS-TENG exhibits stable electrical output over 2000 droplet cycles and retains excellent performance after one month of placement, demonstrating its robust stability. The device remains functional under humid conditions as the leakage current density at ∼90% RH stays within the insulating regime (1.0×10^{-8} A/cm^2). Furthermore, LS-TENG, with its fully coated layers, demonstrates exceptional design flexibility, allowing it to be easily fabricated on various substrates such as PMMA, glass, Al_2O_3, and flexible polyvinyl chloride (Figure 4f). This versatility is confirmed by its consistent electrical output performance across these substrates, as shown in Figure S19. Additionally, the concentration of AgNWs in the spraying process affects both the conductivity and transparency of the conductive coating, thereby influencing the overall electrical output and transparency of the entire LS-TENG. As illustrated in Figure S20, reducing the spraying concentration results in a slight decrease in output voltage and current, while light transmittance increases significantly. Thus, a high-performance LS-TENG with high transparency can be applied to diverse substrates, such as greenhouses and solar panels, without compromising their surface patterns and functionalities, contributing to the establishment of a more comprehensive distributed energy system.

3.3. Applications of LS-TENG. The high-performance LS-TENG, incorporating a PFSC triboelectric and AgNW conductive coating, demonstrates broad application prospects for droplet energy harvesting. As evidenced, the LS-TENG is capable of lighting up 27 LEDs (Figure 5a and Video S1), as well as charging capacitors with capacitances of 1 μF, 2.2 μF, and 4.7 μF through rectification (Figure 5b), and subsequently powering a timer (Figure 5c). These results show the promising potential of LS-TENG for efficiently powering smart electronic devices.

Meanwhile, due to the random and widespread distribution of outdoor raindrops, it is necessary to develop a large-scale LS-TENG to ensure adequate contact with them. As shown in Figure 5d, the interdigital electrodes used in the large-scale LS-TENG can be conveniently fabricated using mask spraying. Each pair of interdigital electrodes functions as a set of double electrodes to achieve electrical output, increasing the possibility of effective contact and improving total electrical output performance. Additionally, this design simplifies the system by requiring only two connection wires, avoiding the complicated external circuit compared to the traditional LS-TENG array, where each unit requires separate wiring connections. To simulate a variety of raindrop distributions, a droplet dispensing system equipped with multiple nozzles is installed. The electrical output performance of the large-scale LS-TENG under different rainfall modes is illustrated in Figure 5e,f. Light rain, moderate rain, and heavy rain are simulated at rates of 500, 1500, and 2500 mL/h, respectively. As the rainfall rate increases, both the frequency and peak values of the output rise. In heavy rain conditions, the maximum output voltage and current reach 21 V and 24 μA, respectively. Given the substantial advantages of coating technology and

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interdigital electrodes, LS-TENG can be efficiently mass-produced with excellent electrical output performance, opening up promising opportunities for harvesting raindrop energy and powering a variety of devices. Furthermore, as presented in Figure S21, the large-scale LS-TENG maintains relatively stable output under light and moderate simulated rainfall, while heavy rainfall induces performance degradation, indicating potential for further optimization.

4. CONCLUSIONS

To summarize, the PFSC composite coating has been successfully fabricated using the doping method and applied in LS-TENG for droplet energy harvesting. The addition of SiO_2@Cu and FOTS simultaneously enhances the dielectric constant and hydrophobicity of the coating through the polarization effect and surface tuning. These improvements contribute to a significant enhancement in the electrical output performance of the LS-TENG. The LS-TENG featuring the PFSC composite coating exhibits outstanding output, generating a voltage of 19 V and a current of 6.7 μA from a single waterdrop, which represents 2.5 and 2.8 times the performance of pure PVDF-HFP, respectively. This device is capable of lighting 27 LEDs and operating a timer. Furthermore, all layers in the LS-TENG, including the PFSC triboelectric layer and the AgNW conductive layer, are achieved through a simple coating process, providing high scalability, flexibility, and transparency to the LS-TENG. Consequently, this method offers an efficient approach to create a high-performance, easily fabricated LS-TENG, which holds promise for diverse applications in harvesting droplet energy to power electronic devices across various fields.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.5c14513.

TEM images of SiO_2 and SiO_2@Cu; XPS spectra of SiO_2@Cu; SEM images and EDS mappings of SiO_2@Cu with different Cu:SiO_2 mass ratios; XRD pattern and FTIR spectrum of PVDF-HFP, PF, PFS, PFC, and PFSC coatings; cross-section SEM image of PFSC coating; SEM images of PFSC coating with different SiO_2@Cu doping concentrations; comparison of recently reported LS-TENG; current−time curves of LS-TENG with PVDF-HFP, PF, PFS, PFC, and PFSC coatings; output performance of LS-TENG at different SiO_2@Cu composition ratios and doping concentrations; output performance of LS-TENG at different FOTS doping concentrations; output performance of LS-TENG at different PFSC drop-casting volumes; theoretical correlation between dielectric constant and electrical output performance; dielectric constant of PVDF-HFP, PF, PFS, PFC, and PFSC coatings at 10^3 Hz; sliding angle and contact angle hysteresis of PVDF-HFP, PF, PFS, PFC, and PFSC coatings; high-speed camera images during dynamic movement of droplets on PVDF-HFP, PF, PFS, PFC, and PFSC coatings; analysis of $\Delta A/\Delta t$ during dynamic movement of droplets on PVDF-HFP, PF, PFS, PFC, and PFSC coatings; output performance of LS-TENG at different inclined angles and dropping heights; Weber number of PFSC coating at different dropping heights; stability of PFSC-based LS-TENG under successive droplet cycles and ambient environmental placement; output performance of LS-TENG on different substrates; output performance and light transmittance of LS-TENG with different AgNW spraying concentrations; and stability of large-scale LS-TENG under different rainfall modes (PDF)

LS-TENG successfully lighting up 27 LEDs (MP4)

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Notes

The authors declare no competing financial interest.

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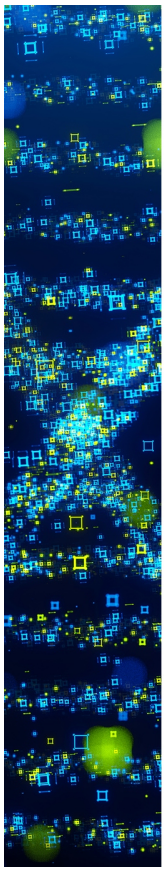
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